The Differential: what the simulation showed

West bay 1: the Differential · 20 October 2026

Meshing

The four 16-tooth mitre gears mesh with each spider turned half a tooth (11.25 degrees) from the side gears, as the geometry says: each side gear has a tooth pointing at each spider, so the spider needs a gap there.

Checked outside Bullet by intersecting the printed meshes (trimesh, manifold3d): the overlap volume is zero between 11.0 and 11.5 degrees, and rises steadily on either side, to 488 mm³ at 0 degrees.

Mesh-to-mesh contact (GImpact) doesn't work for these teeth

With each gear's printed mesh as its collision shape (Bullet's GImpact), the gears never behaved: - **Holding fails:** an input held by its motor was dragged round with the other. - **No clean reading:** the carrier's turn was 10 to 60% off (A + B) / 2. - **Spiders loose:** the spiders spun about axes other than their pins. - **Nothing helped:** changing the margin (0.05 mm), the solver iterations (10 to 50), the motor strength (5 to 500), the friction (0.05 to 0.3), or driving the inputs kinematically. - **Phase search:** looking for the spiders' phase by the kick their teeth gave at rest found only noise: near-zero at isolated angles, large at their neighbours.

Two set-up mistakes were found on the way, then fixed: - An OpenSCAD mesh's place has to be set after it is added to the world. At first all four gears sat at the origin. - A part's collisions are fixed when it is added. Turning `collides` on afterwards does nothing.

Convex pieces do work

Each gear as a compound of convex hulls: its body inside the root cone (224 points), and its 16 teeth outside it (one 24-point hull, placed 16 times). The tooth's hull is only 2% bigger than the tooth, the body's 1.6%. Masses and inertias come from the printed parts at 0.8 g/cm³. With motors on the inputs, over 4 s:

A (turns/s)	B (turns/s)	carrier (turns)	(A + B) / 2	error
0.25	0 (held)	+0.4751	+0.4751	0.00%
0.25	0.25	+0.9495	+0.9495	0.00%
0.25	-0.25	-0.0009	-0.0009	(2e-5 turns)
0.5	0.1	+1.1461	+1.1461	0.00%
-0.4	0.3	-0.1926	-0.1926	0.02%

That was at 10 solver iterations, and it ran faster than real time. The exhibit uses 20.

The hand

- **Both at once:** turning both inputs together and then stopping one jams the simulated teeth. The load jumps across the backlash all at once. So the hand turns one input at a time, as a person turns one crank at a time.
- **Holding:** an input held only at zero speed creeps (0.08 units) under the load from the other. The hand steers the held one back to where it was set.
- **Back-driving:** at friction 0.1 the held input is pushed 0.3 units before the hand corrects it. That is realistic: a slippery differential back-drives its other input unless it is held firmly.

The tour (3 + 4, then 12, then 10, then 0)

Setting	Result
friction 0.1	PASS
friction 0.3	PASS
friction 0.5	PASS
hand speed 0.2 turns/s	PASS
hand speed 0.4 turns/s	PASS
hand speed 0.8 turns/s	PASS

The ring agrees with A + B to 0.01 units every time.

Resting

Once the hand has nothing to do and nothing has moved for half a second, the Differential stops stepping until there is something to do again. Stepped at 2400 steps a second, a machine at rest costs as much as one running, and its parts never quite stop (their sleeping is turned off, as it must be while they work): what was left once it had settled was up to about 0.05 cm/s and 0.03 rad/s (the gears in their backlash). "Not moving" is set just above that: 0.1 cm/s and 0.05 rad/s. Resting can't change a result: woken, it carries on from exactly the state it stopped in, speeds and all. Its sweep (friction 0.1, 0.3, 0.5; hand 0.2 and 0.8 turns a second) was run again with resting on: 6/6. Over a 70 s run, 79 s of computing became 16 s.